Section Overview

Inheritance

- What is Inheritance?
  - Why is it useful?
- Terminology and Notation
- Inheritance vs. Composition
- Deriving classes from existing classes
  - Types of inheritance
- Protected members and class access
- Constructors and Destructors
  - Passing arguments to base class constructors
  - Order of constructor and destructors calls
- Redefining base class methods
- Class Hierarchies
- Multiple Inheritance

Inheritance

What is it and why is it used?

- Provides a method for creating new classes from existing classes
- The new class contains the data and behaviors of the existing class
- Allow for reuse of existing classes
- Allows us to focus on the common attributes among a set of classes
- Allows new classes to modify behaviors of existing classes to make it unique
    - Without actually modifying the original class

Inheritance

Related classes

- Player, Enemy, Level Boss, Hero, Super Player, etc.
- Account, Savings Account, Checking Account, Trust Account, etc.
- Shape, Line, Oval, Circle, Square, etc.
- Person, Employee, Student, Faculty, Staff, Administrator, etc.

Inheritance

Accounts – with inheritance – code reuse

```
class Account {
    // balance, deposit, withdraw, . . .
};

class Savings_Account : public Account {
    // interest rate, specialized withdraw, . . .
};

class Checking_Account : public Account {
    // minimum balance, per check fee, specialized withdraw, . . .
};

class Trust_Account : public Account {
    // interest rate, specialized withdraw, . . .
```

Inheritance

Terminology

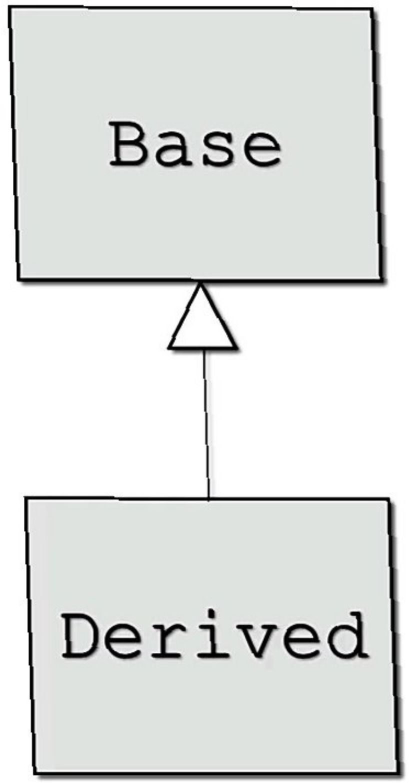
- Inheritance
  - Process of creating new classes from existing classes
  - Reuse mechanism
- Single Inheritance
  - A new class is created from another 'single' class
- Multiple Inheritance
  - A new class is created from two (or more) other classes

Inheritance

Terminology

- Base class (parent class, super class)
  - The class being extended or inherited from

- Derived class (child class, sub class)
  - The class being created from the Base class
  - Will inherit attributes and operations from Base class



Inheritance

Terminology

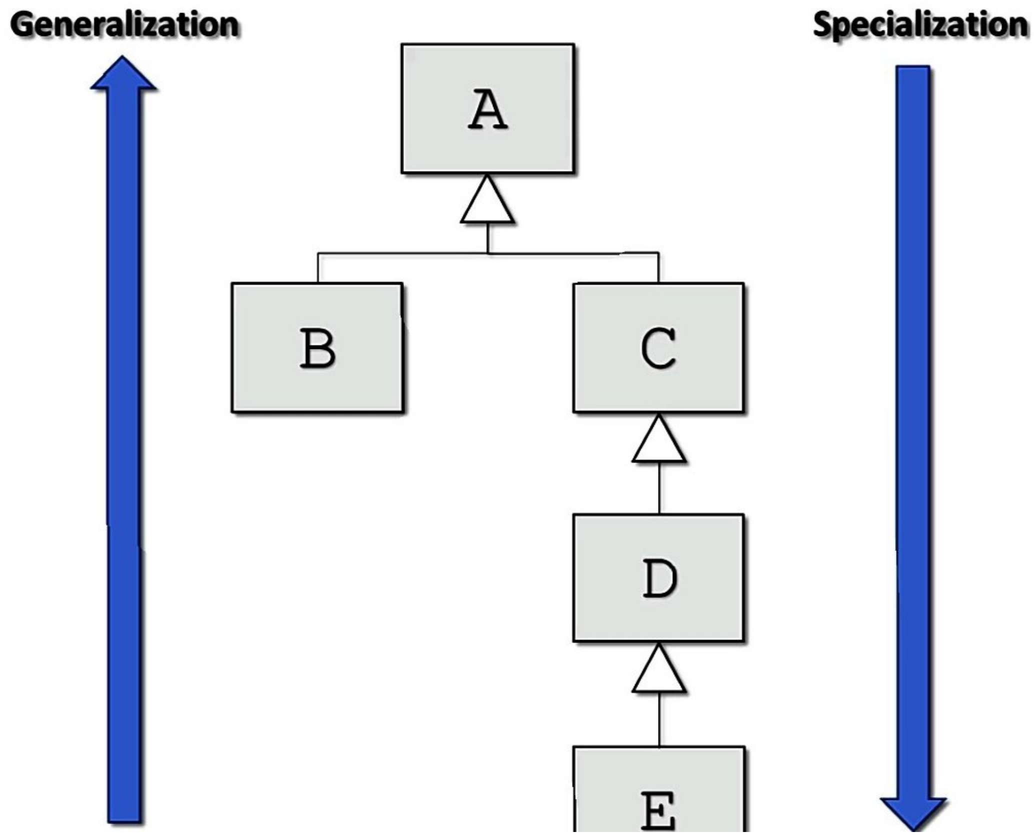
- “Is-A” relationship
  - Public inheritance
  - Derived classes are sub-types of their Base classes
  - Can use a derived class object wherever we use a base class object
- Generalization
  - Combining similar classes into a single, more general class based on common attributes
- Specialization
  - Creating new classes from existing classes proving more specialized attributes or operations
- Inheritance or Class Hierarchies
  - Organization of our inheritance relationships

Inheritance

Class hierarchy

Classes:

- A
- B is derived from A
- C is derived from A
- D is derived from C
- E is derived from D

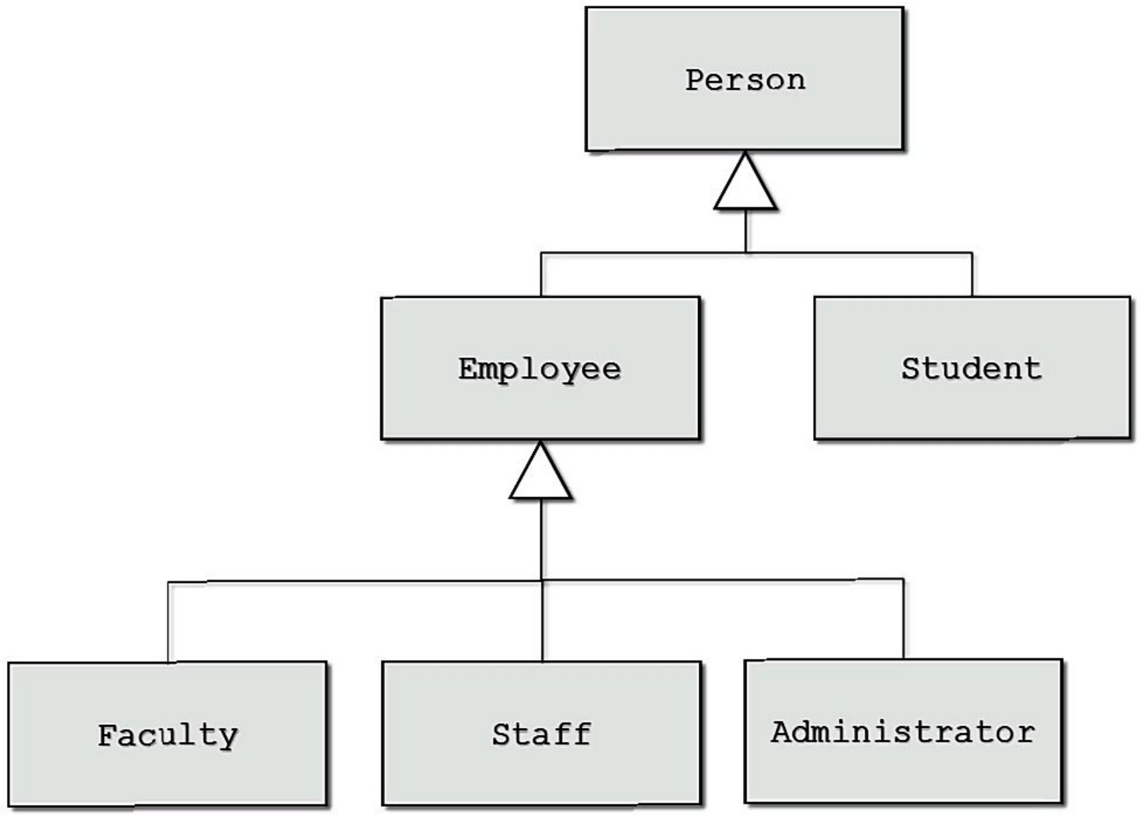


Inheritance

Class hierarchy

Classes:

- Person
- Employee is derived from Person
- Student is derived from Person
- Faculty is derived from Employee
- Staff is derived from Employee
- Administrator is derived from Employee



Inheritance

Public Inheritance vs. Composition

- Both allow reuse of existing classes

- Public Inheritance
  - “is-a” relationship
    - Employee ‘is-a’ Person
    - Checking Account ‘is-a’ Account
    - Circle “is-a” Shape

- Composition
  - “has-a” relationship
    - Person “has a” Account
    - Player “has-a” Special Attack
    - Circle “has-a” Location

Deriving classes from exiting classes

C++ derivation syntax

```
class Base {
  // Base class members . . .
};

class Derived: access-specifier Base {
   // Derived class members . . .
};
```

Access-specifier can be: `public`, `private`, or `protected`

Deriving classes from exiting classes

Types of inheritance in C++

- `public`
  - Most common
  - Establishes ‘**is-a**’ relationship between Derived and Base classes
- `private` and `protected`
  - Establishes “derived class **has a** base class” relationship
  - “Is implemented in terms of” relationship
  - Different from composition

Deriving classes from exiting classes

C++ derivation syntax

```
class Account {
  // Account class members . . .
};

class Savings_Account: public Account {
   // Savings_Account class members . . .
};
```

`Savings_Account` ‘is-a’ `Account`

Deriving classes from exiting classes

C++ creating objects

```
Account account {};
Account *p_account = new Account();

account.deposit(1000.0);
p_account->withdraw(200.0);

delete p_account;
```

Protected Members and Class Access

The protected class member modifier

```
class Base {

  protected:
    // protected Base class members . . .
};
```

- Accessible from the Base class itself
- Accessible from classes Derived from Base
- Not accessible by objects of Base or Derived

Protected Members and Class Access

The protected class member modifier

```
class Base {
   public:
     int a;   // public Base class members . . .


  protected:
     int b;   // protected Base class members . . .


  private:
     int c;   // private Base class members . . .

};
```

Deriving classes from exiting classes

Access with **`public`** inheritance

Base Class		Access in Derived Class
`public: a` `protected: b` `private: c`	**public** inheritance	`public: a` `protected: b` `c :` **`no access`**

Deriving classes from exiting classes

Access with **`private`** inheritance

Base Class		Access in Derived Class
`public: a`	**private** inheritance	**`private`**`: a`
`protected: b`		**`private`**`: b`
`private: c`		`c :` **`no access`**

Constructors and Destructors

Constructors and class initialization

- A Derived class inherits from its Base class

- The Base part of the Derived class MUST be initialized BEFORE the Derived class is initialized

- When a Derived object is created
  - Base class constructor executes then
  - Derived class constructor executes

Constructors and Destructors

Constructors and class initialization

```
class Base {
public:
    Base(){ cout << "Base constructor" << endl; }
};

class Derived : public Base {
public:
    Derived(){ cout << "Derived constructor " << endl; }
};
```

Constructors and Destructors

Constructors and class initialization

	Output
`Base base;`	`Base constructor`
`Derived derived;`	`Base constructor` `Derived constructor`

Constructors and Destructors

Destructors

- Class destructors are invoked in the reverse order as constructors
- The Derived part of the Derived class MUST be destroyed BEFORE the Base class destructor is invoked
- When a Derived object is destroyed
  - Derived class destructor executes then
  - Base class destructor executes
  - Each destructor should free resources allocated in it’s own constructors

Constructors and Destructors

Destructors

```
class Base {
public:
    Base(){ cout << "Base constructor" << endl; }
    ~Base(){ cout << "Base destructor" << endl; }
};

class Derived : public Base {
public:
    Derived(){ cout << "Derived constructor " << endl; }
    ~Derived(){ cout << "Derived destructor " << endl; }
};
```

Constructors and Destructors

Destructors and class initialization

	Output
`Base base;`	Base constructor Base destructor
`Derived derived;`	Base constructor Derived constructor Derived destructor Base destructor

Constructors and Destructors

Constructors and class initialization

- A Derived class does NOT inherit
  - Base class constructors
  - Base class destructor
  - Base class overloaded assignment operators
  - Base class friend functions

- However, the derived class constructors, destructors, and overloaded assignment operators can invoke the base-class versions

- C++11 allows explicit inheritance of base 'non-special' constructors with
  - `using Base::Base;` anywhere in the derived class declaration
  - Lots of rules involved, often better to define constructors yourself

Inheritance

Passing arguments to base class constructors

- The Base part of a Derived class must be initialized first
- How can we control exactly which Base class constructor is used during initialization?
- We can invoke the whichever Base class constructor we wish in the initialization list of the Derived class

Inheritance

Passing arguments to base class constructors

```
class Base {
public:
    Base();
    Base(int);
    . . .
};

Derived::Derived(int x)
    : Base(x), {optional initializers for Derived} {
    // code
}
```

Constructors and Destructors

Constructors and class initialization

```
class Derived : public Base {
   int doubled_value;
public:
   Derived(): Base{}, doubled_value{0} {
      cout << "Derived no-args constructor " << endl;
   }
   Derived(int x) : Base{x}, doubled_value {x*2} {
      cout << "int Derived constructor " << endl;
   }
};
```

Constructors and Destructors

Constructors and class initialization

```
class Derived : public Base {
   int doubled_value;
public:
   Derived(): Base{}, doubled_value{0} {
      cout << "Derived no-args constructor " << endl;
   }
   Derived(int x) : Base{x}, doubled_value {x*2} {
      cout << "int Derived constructor " << endl;
   }
};
```

Constructors and Destructors

Constructors and class initialization

	Output
`Base base;`	`Base no-args constructor`
`Base base{100};`	`int Base constructor`
`Derived derived;`	`Base no-args constructor` `Derived no-args constructor`
`Derived derived{100};`	`int Base constructor` `int Derived constructor`

Inheritance

Copy/Move constructors and overloaded operator=

- Not inherited from the Base class
- You may not need to provide your own
    - Compiler-provided versions may be just fine
- We can explicitly invoke the Base class versions from the Derived class

Inheritance

Copy constructor

- Can invoke Base copy constructor explicitly
  - Derived object *'other'* will be **sliced**

```
Derived::Derived(const Derived &other)
    : Base(other), {Derived initialization list}
{
    // code
}
```

Constructors and Destructors

Copy Constructors

```
class Base {
   int value;
public:
   // Same constructors as previous example

   Base(const Base &other) :value{other.value} {
   cout << "Base copy constructor" << endl;
   }
};
```

Constructors and Destructors

operator=

```
class Base {
    int value;
public:
    // Same constructors as previous example
    Base &operator=(const Base &rhs) {
        if (this != &rhs) {
            value = rhs.value;   // assign
        }
        return *this;
    }
};
```

Constructors and Destructors

operator=

```
class Derived : public Base {
   int doubled_value;
public:
   // Same constructors as previous example
   Derived &operator=(const Derived &rhs) {
      if (this != &rhs) {
         Base::operator=(rhs);          // Assign Base part
         doubled_value = rhs.doubled_value; // Assign Derived part
      }
      return *this;
   }
};
```

Inheritance

Copy/Move constructors and overloaded operator=

- Often you do not need to provide your own

- If you **DO NOT** not define them in Derived
  - then the compiler will create them and automatically and call the base class's version

- If you **DO** provide Derived versions
  - then **YOU** must invoke the Base versions **explicitly** yourself

- Be careful with raw pointers
  - Especially if Base and Derived each have raw pointers
  - Provide them with deep copy semantics

Inheritance

Using and redefining Base class methods

- Derived class can directly invoke Base class methods
- Derived class can **override** or **redefine** Base class methods
- Very powerful in the context of polymorphism
  (next section)

Inheritance

Using and redefining Base class methods

```
class Account {
public:
   void deposit(double amount) { balance += amount; }
};

class Savings_Account: public Account {
public:
   void deposit(double amount) { // Redefine Base class method
      amount += some_interest;
      Account::deposit(amount);  // invoke call Base class method
   }
};
```

Inheritance

Static binding of method calls

- Binding of which method to use is done at compile time
  - Default binding for C++ is static
  - Derived class objects will use Derived::deposit
  - But, we can explicitly invoke Base::deposit from Derived::deposit
  - OK, but limited – much more powerful approach is dynamic binding which we will see in the next section

Inheritance

Static binding of method calls

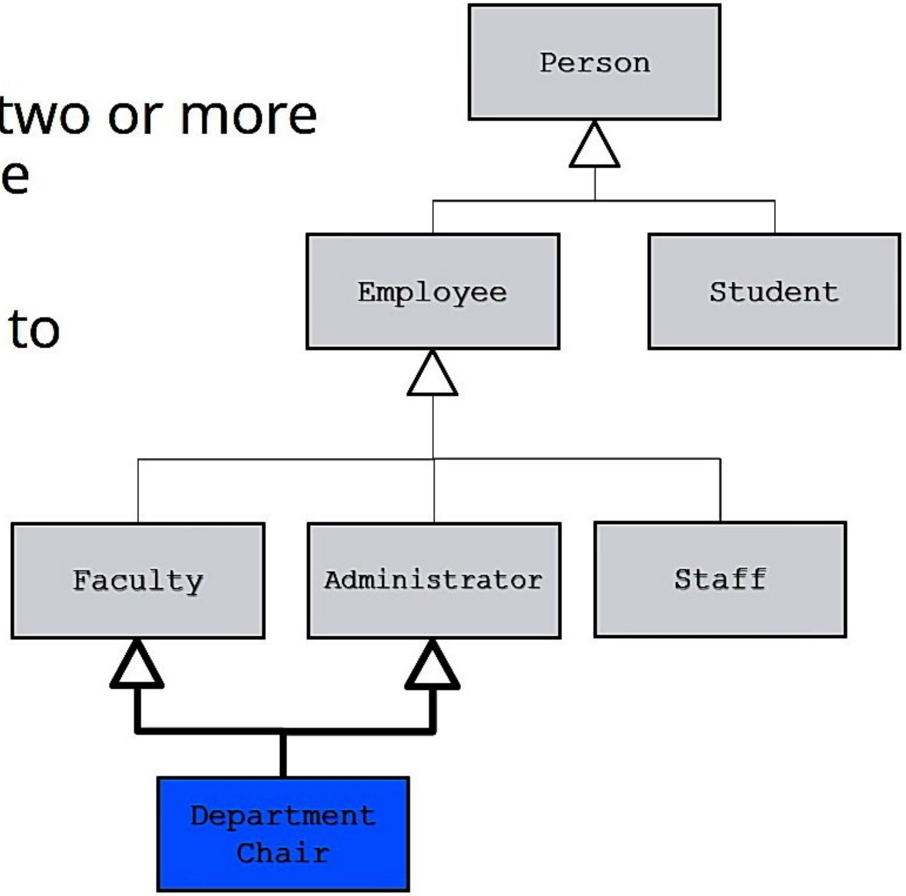
```
Base b;
b.deposit(1000.0);                  // Base::deposit

Derived d;
d.deposit(1000.0);                  // Derived::deposit

Base *ptr = new Derived();
ptr->deposit(1000.0);               // Base::deposit ????
```

Multiple Inheritance

- A derived class inherits from two or more Base classes at the same time
- The Base classes may belong to unrelated class hierarchies
- A Department Chair
  - Is-A Faculty and
  - Is-A Administrator



Multiple Inheritance

C++ Syntax

```
class Department_Chair:
        public Faculty, public Administrator {
    . . .
};
```

- Beyond the scope of this course
- Some compelling use-cases
- Easily misused
- Can be very complex